

MS 6015: Multivariate Statistical Methods for Business

Probability Review ¹

¹Reference: A First Course in Probability: Sheldon Ross     1 / 73 1 / 73

Outline

Combinatorial Analysis

- Principle of Counting

Axioms of Probability

- Random Experiment, Sample Spaces and Events

- Conditional Probability and Independent Events

Random Variables

- Discrete Random Variables

- Continuous Random Variables

Jointly distributed Random Variables

- Jointly Distributed Discrete Random variables

- Jointly Distributed Continuous Random variables

- Independent Random Variables

- Sums of Independent Random Variables

- Conditional Distributions

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- ▶ Basic Principle of Counting: Suppose that two experiments are being performed. If experiment 1 can result in m outcomes, and experiment 2 results in n outcomes. Then together there are mn possible outcomes for the two experiments.
- ▶ Generalized Principle of Counting: If r experiments are performed with i^{th} experiment having n_i outcomes, then there is a total of $n_1 \cdot n_2 \cdot \dots \cdot n_r$ possible outcomes of the r experiments

Permuations

- ▶ A permutation is an arrangement of a group of objects in a particular order
- ▶ The total number of permutations that can be formed from n objects using all of them without repetition is $n! = n.(n - 1) \dots 1$.
- ▶ symbol $n!$ is read n factorial. By definition, $0! = 1$
- ▶ The total number of permutations of n objects of which n_1 are alike , n_2 are alike , ..., n_r are alike $= \frac{n!}{n_1!n_2!\dots n_r!}$

Combinations

- ▶ A combination is a group of objects in which the order or arrangement of the objects is disregarded
- ▶ The number of ways of choosing r objects from n objects is a combination. We define $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.
- ▶ A useful combinatorial identity:

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$

- ▶ $\binom{n}{r}$ is usually referred to a binomial coefficients

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Basic Definitions

Definition

Random Experiment: A random experiment is a process with a set of well defined outcomes, however, the outcome of any given process cannot be determined in advance.

Probability is used to describe the likelihood of a certain outcome from a random experiment. Probability theory is usually expressed in terms of sets.

Definition

Sample Space: The set of all possible outcomes of a random experiment is defined as the sample space of the experiment and is denoted by S .

Definition

Event: Any subset E of the sample space is known as an event. Events are usually denoted by capital letters, A, B, C, E, F, G , etc.

General interest is to find probability of a event

An event will occur if any of the outcomes in that subset occurs.

Some important operations on events.

Definition

A and B are two events in the sample space S . We can define an event C that both the events A and B occur. In terms of sets, C is the intersection of the two sets A and B , this is written as $C = A \cap B$.

Definition

A and B are two events in the sample space S . We can define an event D that at least one of the events A and B occur. In terms of sets, C is the union of the two sets A and B , this is written as $C = A \cup B$

Definition

A is an event in the sample space S . We can define an event E that A does not occur. In terms of sets, E is the complement of A and, this is written as $E = A^c$

Definition

Two events A and B are mutually exclusive if they cannot occur at the same time. We say that $A \cap B = \phi$, where ϕ is the null set.

DeMorgan's Laws

$$\left(\bigcup_{i=1}^{\infty} E_i \right)^c = \bigcap_{i=1}^{\infty} E_i^c$$
$$\left(\bigcap_{i=1}^{\infty} E_i \right)^c = \bigcup_{i=1}^{\infty} E_i^c$$

Interpreting Probability

There are two possible ways to interpret probabilities:

- ▶ **Frequency interpretation:** Under the frequency interpretation, probability of an event is defined as the (limiting) proportion of times the event occurs.
 - ▶ If $n(E)$ is the number of times an event E is repeated in n trials of the experiment, then probability of the event E is defined as $P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$.
- ▶ **Axiomatic Approach :** In this approach we assume for each event E , in the sample space S , there is a value (or a set function) $P(E)$, the value is referred to as probability of E . These probabilities satisfy certain axioms. They are

Axioms of Probability

1. Non negativity: $0 \leq P(E) \leq 1$
2. Total one: $P(S) = 1$
3. For any sequence of mutually exclusive events, $E_1, E_2, \dots$, we have $P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$

- └ Axioms of Probability
- └ Conditional Probability and Independent Events

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Conditional Probability

Definition

Suppose we know that an event F has occurred. In light of this information, we must revise the probability of other events E .

The revised probability is called the conditional probability of E given F and is defined by

$$\text{If } P(F) > 0, \text{ then } P(E | F) = \frac{P(E \cap F)}{P(F)}$$

Bayes Theorem

Let E and F be events . We may express $E = EF \cup EF^c$, a union of two mutually exclusive events..Then

$$\begin{aligned} P(E) &= P(EF \cup EF^c) \\ &= P(EF) \cup P(EF^c) \\ &= P(E | F) P(F) + P(E | F^c) P(F^c) \end{aligned}$$

Generalization: Let $F_1, F_2, \dots, F_n$ be a partition of the Sample space S into pairwise mutually exclusive (i.e. disjoint) events. Then for any event E ,

$$P(E) = P(E | F_1) P(F_1) + P(E | F_2) P(F_2) + \dots + P(E | F_n) P(F_n)$$

Bayes Formula

Bayes Formula: Let $F_1, F_2, \dots, F_n$ be a partition of the Sample space S into pairwise mutually exclusive (i.e. disjoint) events. Then for any event E ,

$$P(F_j | E) = \frac{P(E | F_j) P(F_j)}{P(E | F_1) P(F_1) + P(E | F_2) P(F_2) + \dots + P(E | F_n) P(F_n)}$$

Independent Events

Definition

Two events E and F are said to be independent if $P(E \cap F) = P(E) P(F)$.

Definition

Three events E , F , and G are said to be independent if

$$P(EFG) = P(E) P(F) P(G)$$

$$P(EF) = P(E) P(F)$$

$$P(EG) = P(E) P(G)$$

$$P(FG) = P(F) P(G)$$

Definition

Random Variable, rv : a real-valued function defined on the sample space. That is, a random variable is a function that maps outcomes in the sample space to the real numbers.

Definition

Probability Distribution Function, pdf, $f(X)$: the function which gives the probability of a random variable at different intervals or at specified real numbers across its range.

Definition

Cumulative Distribution Function, cdf, $F(X)$: a distribution function that gives the probability of the random variable X being less than a specified real number. $F(a) = P(X \leq a)$

Expectation and Variance

Definition

Expected Value, $E(X)$, $\mu(X)$, is a function of a random variable that gives a measure of its central location. It is the probability weighted average of the possible values that X can take on.

Definition

Variance, $Var(X)$, $\sigma^2(X)$ the square of the expected value of the difference of X and $E(X)$.

Definition

Standard Deviation, $SD(X)$, $\sigma(X)$, the square root of the variance.

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Definition

Discrete Random Variable: a random variable that can take on at most a countable number of possible values.

Definition

Probability Mass Function, pmf, $p_X(X)$: the distribution function of a discrete random variable. The probability mass function (pmf) of a discrete random variable is given by

$p_X(X) = P(X = x)$. We have $\sum_{all\ x} P(X = x) = 1$.

Special Discrete Distributions

DISTRIBUTION	DESCRIPTION	Parameters
Bernoulli	Single trial with two outcomes	p
Binomial	Count number of successes in n trials	n, p
Poisson	n very large, p very small . Ex: Count number of misprints in a book	$\lambda = np$
Geometric	Number of trials for the <i>first</i> success	p
Negative Binomial	Number of trials for r successes	r, p
Hypergeometric	In a sample of n , Number of white balls selected from m white and $N - m$ black balls	N, m, n

DISTRIBUTION	Parameters	pmf $P(X = i)$	Range of i	$E(X)$	$Var(X)$
Bernoulli	p	$= \begin{cases} 1 & \text{if success} \\ 0 & \text{otherwise} \end{cases}$	$i = 0, 1$	p	$p(1 - p)$
Binomial	n, p	$\binom{n}{i} p^i (1 - p)^{n-i}$	$i = 0, 1, \dots, n$	np	$np(1 - p)$
Poisson	$\lambda = np$	$\frac{e^{-\lambda} \lambda^i}{i!}$	$i = 0, 1, 2, \dots$	λ	λ
Geometric	p	$(1 - p)^{i-1} p$	$i = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Bino- mial	r, p	$\binom{n-1}{r-1} p^r (1 - p)^{n-r}$	$i = r, r+1, \dots$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Hypergeometric	N, m, n	$\frac{\binom{m}{i} \binom{N-m}{n-i}}{\binom{N}{n}}$	$i = 0, 1, \dots, n$	$\frac{nm}{N}$	$\frac{N-n}{N-1} np(1 - p)$

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Continuous Random Variables

Definition

X is a continuous random variable if there exists a non negative function f defined for all real $x \in (-\infty, \infty)$ having the property that for any set B of real numbers

$$P(X \in B) = \int_B f(x) dx$$

Definition

$f(x)$ is called the probability density function (pdf) of X

DISTRIBUTION	Parameters	pdf $f(x) =$	Range of x	$E(X)$	$Var(X)$
Uniform	a, b	$\begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$	$a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential	λ	$\begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	μ, σ^2	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$-\infty < x < \infty$	μ	σ^2

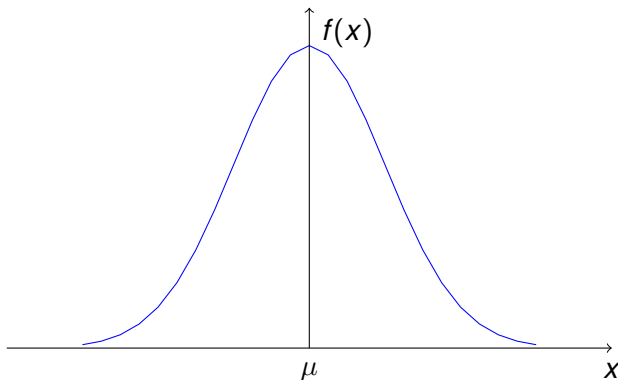
pdf of a Normal distribution

- ▶ The probability density function of a normal random variable X is determined by two parameters:
 - ▶ the expected value of X , denoted by μ
 - ▶ the variance of X , denoted by σ^2 .
- ▶ We write $X \sim N(\mu, \sigma^2)$
- ▶ The Density function is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < \infty$$

Shape of Normal pdf

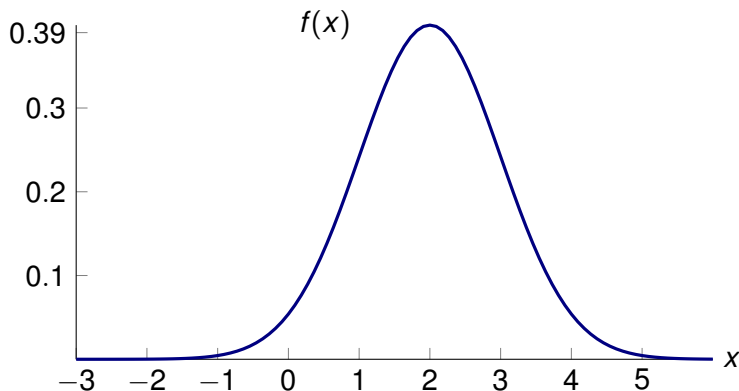
- ▶ The normal probability density function is a bell-shaped density curve that is symmetric about the value μ .
- ▶ Its variability is measured by standard deviation σ .



Height of curve

The height of the curve above point x on the abscissa is

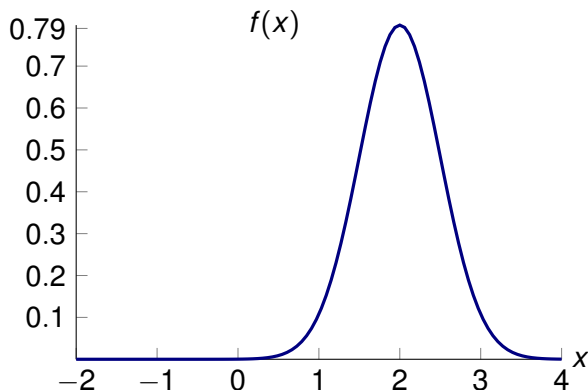
$$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}; \mu = 2, \sigma = 1, \text{ height} = 0.399$$



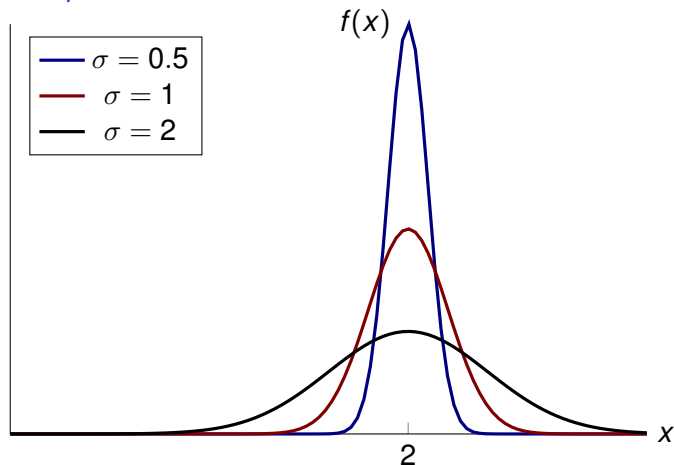
Height of curve

The height of the curve above point x on the abscissa is

$$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}; \mu = 2, \sigma = 0.5, \text{ height} = 0.798$$

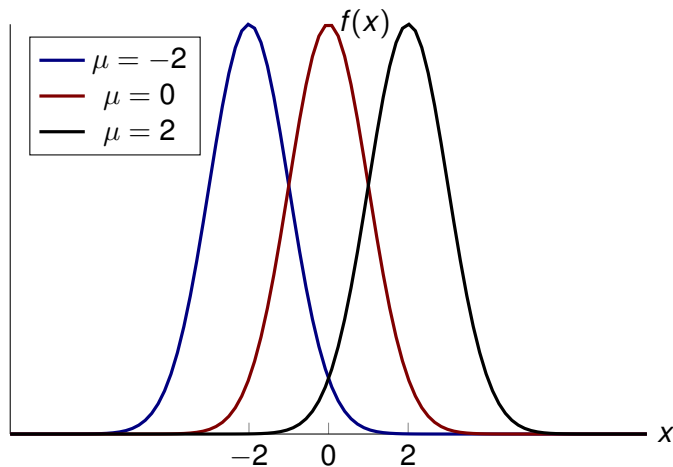


Same μ different σ



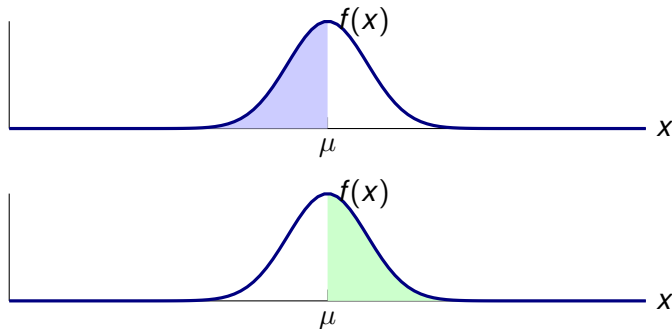
The larger the σ , the flatter and more spread out is the distribution.

Same σ different μ



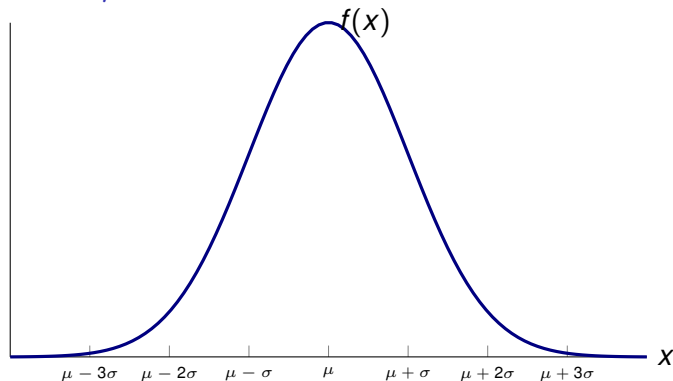
Symmetric around mean

The probability density function of a normal random variable X is symmetric about its expected value μ , it follows that X is equally likely to be on either side of μ . That is,



$$P(X \leq \mu) = P(X \geq \mu) = \frac{1}{2}$$

General μ and σ

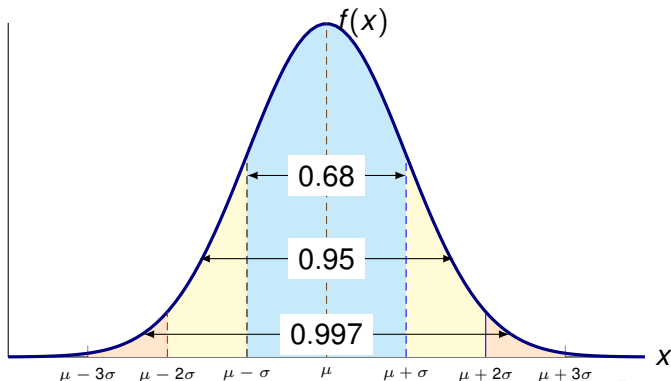


1. bell shaped,
2. centered at μ and
3. close to the horizontal axis outside the range from $\mu - 3\sigma$ to $\mu + 3\sigma$

Approximation rule

A normal random variable with mean μ and standard deviation σ will be

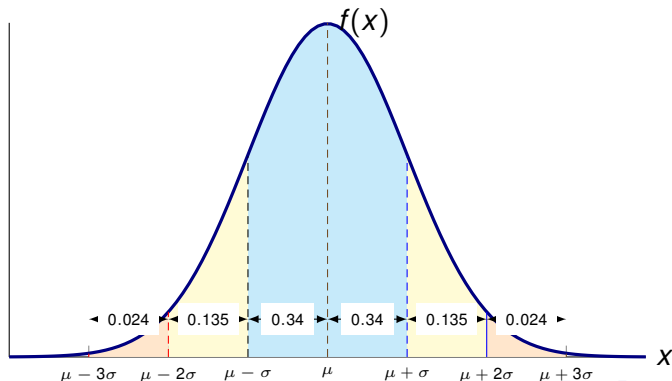
- ▶ Between $\mu - \sigma$ and $\mu + \sigma$ with approx. probability 0.68
- ▶ Between $\mu - 2\sigma$ and $\mu + 2\sigma$ with approx. probability 0.95
- ▶ Between $\mu - 3\sigma$ and $\mu + 3\sigma$ with approx. probability 0.997



Approximation rule

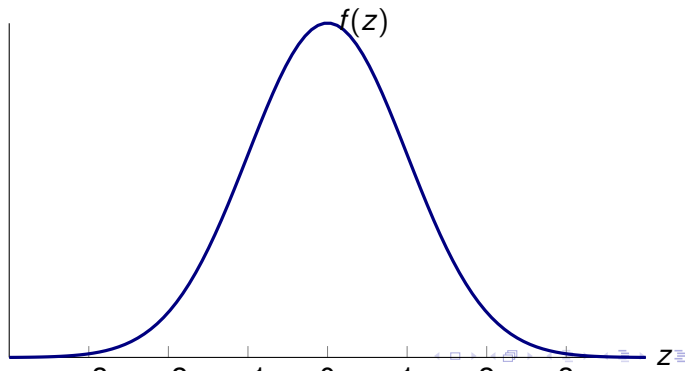
A normal random variable with mean μ and standard deviation σ will be

- ▶ Between $\mu - \sigma$ and $\mu + \sigma$ with approx. probability 0.68
- ▶ Between $\mu - 2\sigma$ and $\mu + 2\sigma$ with approx. probability 0.95
- ▶ Between $\mu - 3\sigma$ and $\mu + 3\sigma$ with approx. probability 0.997



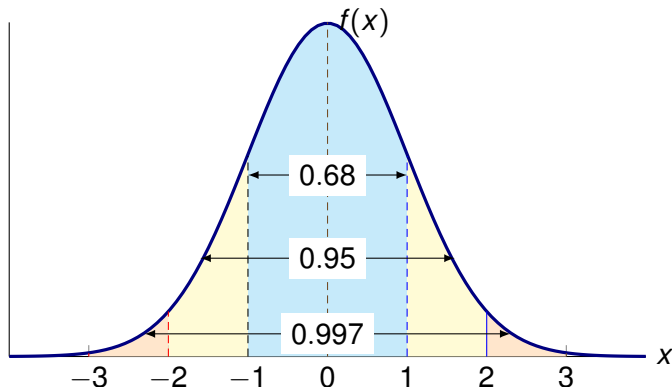
Standard Normal Distribution: $\mu = 0$ and $\sigma = 1$

- ▶ A normal random variable having mean value 0 and standard deviation 1 is called a standard normal random variable, and its density curve is called the standard normal curve.
- ▶ We represent a standard normal random variable by Z .



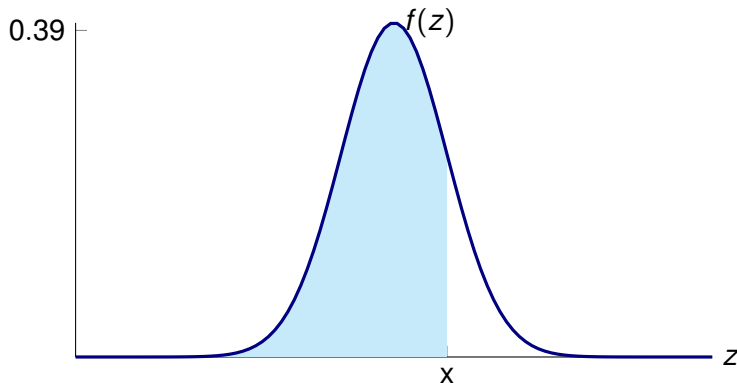
Approximation rule for standard normal

- ▶ Between -1 and $+1$ with approx. probability 0.68
- ▶ Between -2 and $+2$ with approx. probability 0.95
- ▶ Between -3 and $+3$ with approx. probability 0.997



Standard Normal table

- For each nonnegative value of x , the table specifies the probability that Z is less than x , i.e. $P(Z \leq x)$.



Portion of Standard Normal table

<i>z</i>	0	0.01	0.02	0.03	0.04	0.05	0.0
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5
0.2	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6
0.4	0.6554	0.6591	0.6628	0.6664	0.67	0.6736	0.6
0.5	0.6915	0.695	0.6985	0.7019	0.7054	0.7088	0.7
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7
0.7	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7
0.8	0.7881	0.791	0.7939	0.7967	0.7995	0.8023	0.8
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8

Standard Normal table

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.648	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.67	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.695	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.719	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.791	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.937	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.975	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.983	0.9834	0.9838	0.9842	0.9846	0.985	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.992	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.994	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.996	0.9961	0.9962	0.9963	0.9964

Using Normal tables to compute probabilities

1. $P(Z < 2.2) = 0.9861$

<i>z</i>	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890

2. $P(Z > 1.1) = 0.1357$

<i>z</i>	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883

3. $P(0 < Z < 2) = 0.4772$

<i>z</i>	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015

Using Normal tables to compute probabilities

4. $P(-0.9 < Z < 1.2) = 0.7008$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015

$$\begin{aligned}
 P(-0.9 < Z < 1.2) &= P(Z < 1.2) - P(Z < -0.9) = \\
 &P(Z < 1.2) - P(Z > 0.9) = \\
 &P(Z < 1.2) + P(Z < 0.9) - 1 = 0.8849 + 0.8159 - 1 = \\
 &0.7008
 \end{aligned}$$

Using Normal tables to compute probabilities

5. $P(Z > -1.96) = 0.9750$

<i>z</i>	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.975	0.9756	0.9761	0.9767

6. $P(Z < -0.72) = 0.2358$

<i>z</i>	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.7	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7853

Using Normal tables to compute probabilities

7. $P(|Z| < 1.64) = 0.899$

8. $P(|Z| > 1.20) = 0.2302$

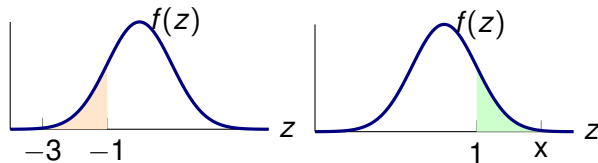
9. $P(-2.2 < Z < 1.2) = 0.8710$

<i>z</i>	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

Using Normal tables to compute unknown x

Find the value of x :

$$P(-3 < Z < -1) = P(1 < Z < x)$$



solution: $x = 3$

Examples

Find value of x

1. $P(Z > x) = 0.05$, $x = 1.65$
2. $P(Z > x) = 0.005$, $x = 2.58$
3. $P(|Z| < x) = 0.75$, $2P(Z < x) = 1.75$; $x = 1.15$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
2.5	0.9938	0.994	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952

Standardizing to Find Normal Probabilities

- ▶ Start by converting x into a z -score

$$Z = \frac{X - \mu}{\sigma}$$

The value of the standardized variable tells us how many standard deviations the original variable is from its mean.

- ▶ We can compute any probability statement about X by writing an equivalent statement in terms of Z

$$P(X < a) = P\left(\frac{X - \mu}{\sigma} < \frac{a - \mu}{\sigma}\right) = P\left(Z < \frac{a - \mu}{\sigma}\right)$$

Additive Property

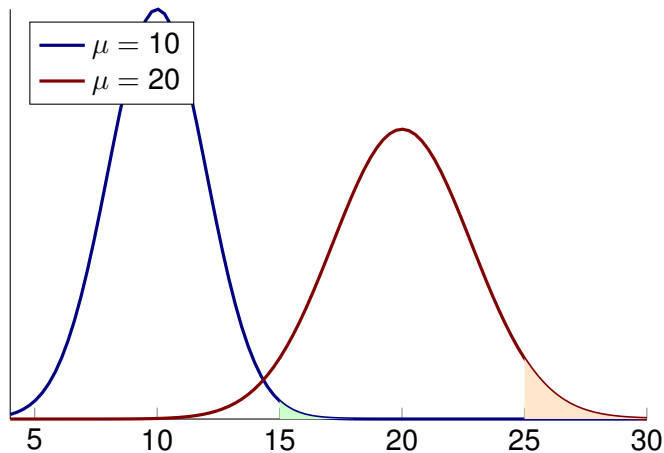
Result

The sum of independent normal random variables is also a normal random variable. Suppose X and Y are independent normal random variables with respective parameters μ_X, σ_X and μ_Y, σ_Y , then $X + Y$ also will be normal with mean $\mu_X + \mu_Y$ and standard deviation $\sqrt{\sigma_X^2 + \sigma_Y^2}$.

Comparing distributions

Let X_1 and X_2 be independent normal random variables, each having mean 10 and variance 4. Which probability is larger

1. $P(X_1 > 15)$
2. $P(X_1 + X_2 > 25)$
3. $P(X_1 + X_2 > 30)$



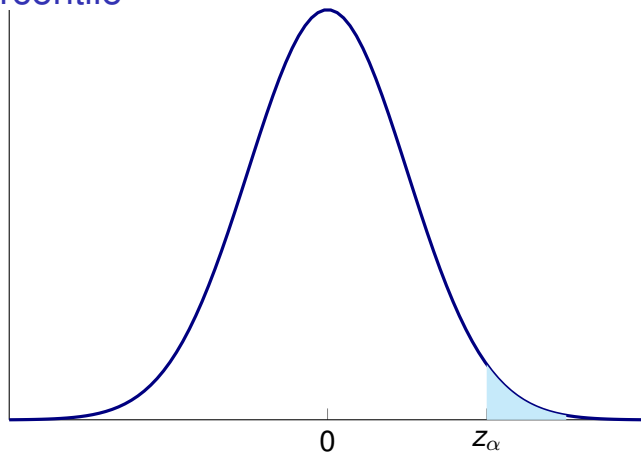
Percentiles

- ▶ The previous problems ask us to find the probability associated with being larger or smaller than some value.
- ▶ Some problems require working in the opposite direction that starts with the probability
- ▶ For any α between 0 and 1, we define z_α to be that value for which

$$P(Z > z_\alpha) = \alpha$$

- ▶ z_α is the $100(1 - \alpha)$ percentile of the standard normal distribution.

Percentile



$$P(Z > z_\alpha) = \alpha$$

$z_{0.025}$

- What is $z_{0.025}$? $P(Z < z_{0.025}) = 1 - P(Z > z_{0.025}) = 0.975$.

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.975	0.9756	0.9761	0.9767

- Search for entry 0.975 in table and find value of x corresponding to this value. We see that $z_{0.025} = 1.96$
- That is, 2.5 percent of the time a standard normal variable will exceed 1.96.
- Since 97.5 percent of the time a standard normal random variable will be less than 1.96, we say that 1.96 is the 97.5 percentile of the standard normal distribution.
- In general, since $100(1 - \alpha)$ percent of the time a standard normal random variable will be less than z_α , we call z_α the $100(1 - \alpha)$ percentile of the standard normal distribution.

Jointly distributed Random Variables

Thus far, we have focused on probability distributions for single random variables. However, we are often interested in probability statements concerning two or more random variables.

- ▶ The joint behavior of two random variables X and Y is determined by the joint cumulative distribution function (cdf):

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

where X and Y are continuous or discrete.

- ▶ In general, if are jointly distributed random variables, the joint cdf is

$$F(x_1, x_2, \dots, x_p) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_p \leq x_p)$$

Principle of Counting

Random Experiment, Sample Spaces and Events

Conditional Probability and Independent Events

Discrete Random Variables

Continuous Random Variables

Jointly Distributed Discrete Random variables

Jointly Distributed Continuous Random variables

Independent Random Variables

Sums of Independent Random Variables

Conditional Distributions

Joint probability mass functions

Joint probability mass functions: Let X and Y be discrete random variables defined on the sample space that take on values $x_1, x_2, \dots$ and $y_1, y_2, \dots$ respectively. The joint probability mass function of (X, Y) is

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Example

- ▶ Suppose that 3 balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. If we let X and Y denote, respectively, the number of red and white balls chosen. Find the joint probability mass function of X and Y , $p(i, j) = P(X = i, Y = j)$.

Example

- Suppose that 3 balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. If we let X and Y denote, respectively, the number of red and white balls chosen. Find the joint probability mass function of X and Y , $p(i, j) = P(X = i, Y = j)$.

$P\{X = i, Y = j\}$					
$i \backslash j$	0	1	2	3	Row sum = $P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column sum = $P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

Marginal Probability Mass Function

- ▶ Suppose that we wish to find the pmf of X and Y from the joint pmf of X and Y .
- ▶ $p_x(x) = \sum_i p(x, y_i)$ and $p_y(y) = \sum_i p(x_i, y)$
- ▶ Case of several variables: If $X_1, X_2, \dots, X_p$ are discrete random variables defined on the sample space, their joint pmf is

$$p(x_1, x_2, \dots, x_p) = P(X_1 = x_1, X_2 = x_2, \dots, X_p = x_p)$$

- ▶ The marginal pmf of X_1 , for example, is

$$p_{X_1}(x_1) = \sum_{x_2, \dots, x_p} p(x_1, x_2, \dots, x_p)$$

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Joint pdf and Joint CDF

- ▶ Suppose that X and Y are continuous random variables. The joint probability density function (pdf) of X and Y is the function $f(x, y)$ such that for every set of pairs of real numbers

$$P((X, Y) \in C) = \int \int_{(x,y) \in C} f(x, y) dx dy$$

- ▶ The joint CDF of (X, Y) is

$$F(a, b) = \int_{-\infty}^b \int_{-\infty}^a f(x, y) dx dy$$

Marginal Pdf

- ▶ The cdf and pdf of can be obtained from the pdf of (X, Y) :

$$P(X \leq x) = \int_{-\infty}^x \int_{-\infty}^{\infty} f(x, y) dy dx$$



$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

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- ▶ For discrete random variables, the condition of independence is equivalent to

$$p(x, y) = p_X(x)p_Y(y) \text{ for all } x, y$$

- ▶ For continuous random variables, the condition of independence is equivalent to

$$f(x, y) = f_X(x)f_Y(y) \text{ for all } x, y$$

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Distribution of $X + Y$ and when X and Y are independent.

► **Sum of Discrete Independent Random Variables:**

Suppose that X and Y are discrete independent random variables. Let $Z = X + Y$. The probability mass function of Z is

$$P(Z = z) = \sum_x P(X = x, Y = z - x) = \sum_x P(X = x)P(Y = z - x)$$

Sums of Independent Random Variables

Distribution of $X + Y$ and when X and Y are independent.

► **Sum of Continuous Independent Random Variables:**

Suppose that X and Y are continuous independent random variables. Let $Z = X + Y$. The CDF of Z is

$$F_Z(a) = \int_{-\infty}^{\infty} F_X(a - y)f_Y(y)dy$$



$$f_Z(a) = \int_{-\infty}^{\infty} f_X(a - y)f_Y(y)dy$$

Sum of independent normal random variables

If $X_i, i = 1, \dots, p$ are independent random variables that are normally distributed with respective means and variances $\mu_i, \sigma_i^2; i = 1, \dots, p$ then $\sum_{i=1}^p X_i$ is normally distributed with mean $\sum_{i=1}^p \mu_i$ and variance $\sum_{i=1}^p \sigma_i^2$.

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Conditional Distributions: The Discrete Case

- ▶ Suppose X and Y are discrete random variables. The conditional probability mass function of X given $Y = y_j$ is the conditional probability distribution of X given $Y = y_j$. The conditional probability mass function of $X|Y$ is

$$p_{X|Y}(x_i|y_j) = P(X = x_i|Y = y_j) = \frac{p_{X,Y}(x_i, y_j)}{p_Y(y_j)}$$

assuming $p_Y(y_j) > 0$

- ▶ If X and Y are independent random variables, then the conditional probability mass function is the same as the unconditional one.

Conditional Distributions: The Continuous Case

- ▶ If X and Y have a joint pdf $f(x, y)$, the conditional pdf of X given $Y = y$ defined for all values of y such that $f_Y(y) > 0$ by

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

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